

Tested on one shared quantity: what adversarial tests of consciousness can and cannot yet measure

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Running head. Tested on one shared quantity

Abstract

The first large preregistered adversarial test in consciousness science produced no winner. The Cogitate consortium challenged central predictions of global neuronal workspace theory and integrated information theory without adjudicating between them, noting that the challenges extend to any theory sharing those predictions. The outcome is structural: theories are compared almost exclusively on where and when conscious contents are encoded, so one design tests many and discriminates none. A purposive audit coded the stated predictions of a set of theories against eight measurable quantities and classified the content manipulated in 36 studies these theories are tested by or built on. Three kinds of column differ. A shared column, the cortical signature of access, carries convergent predictions from most coded theories; it is what Cogitate tested. A contested column, autonomic channel involvement, carries opposed predictions — supramodular interaction theory denies what interoceptive-inference accounts assert — but cannot yet be run, because no report-independent measure of access to interoceptive content exists. The number and incompatibility of competing action policies are single-occupant columns, predicted by one theory and stated by no other; no reclassification of borderline cells makes either contested. Organism-wide cortical-autonomic coherence is thin rather than empty: no coded theory states a prediction over it, three are compatible with one, and heartbeat-evoked responses supply the nearest single-channel operationalisation. Among theory-addressed studies neutral visual content dominates; no coded theory claims the valenced or interoceptive literature. The experiment that can be run now tests a single-occupant prediction; its informative outcome is the null.

Keywords: consciousness; adversarial collaboration; conscious access; action-policy competition; interoception; heartbeat-evoked response; theory audit

1. Introduction

For thirty years the standard complaint about consciousness science was that its theories were not tested. The complaint has been answered, and the answer was worse. The Cogitate consortium ran a preregistered adversarial test of global neuronal workspace theory (GNWT) and integrated informa-

tion theory (IIT) across fMRI, MEG and intracranial EEG in 256 participants, with pass and fail criteria agreed in advance by proponents of both theories and by a theory-neutral consortium [1], and reported that neither central prediction survived intact [2]. GNWT was challenged by the absence of prefrontal ignition at stimulus offset — a null that cannot be blamed on insensitivity, since offset responses were robust elsewhere and the same prefrontal sites showed strong onset responses to the same stimuli. IIT was challenged by the absence of sustained within-posterior synchronisation. The consortium declined a verdict, adopting a Lakatosian soft-falsification frame, and GNWT proponents have published their own reading of the same data [3].

The usual lesson drawn is about the theories: too vague, too unfalsifiable, too well defended by auxiliary assumptions. The more consequential lesson is about the measurements, and the consortium points at it: the prefrontal prediction is shared by those higher-order theories that treat prefrontal cortex as a supplier of visual content rather than merely as what makes content conscious, and the posterior prediction is shared by recurrent processing theory. An adversarial test, on this reading, does not punish a theory; it punishes a *class of prediction*. A theory whose distinctive prediction happens to be the one everyone else also makes cannot be distinguished by the experiment that tests it, however well that experiment is run.

That is the thesis of this review. The problem is not that theories of consciousness are untested; it is that they predict the same quantity and are therefore tested by the same experiment. If the diagnosis is right, the remedy is not another adversarial collaboration on the same dependent variable. But the remedy is not simply a quantity the theories do not predict, and two senses in which a measurable quantity can be a gap have to be kept apart, because they license different inferences. A column is *contested* when at least two live theories state opposed predictions over it; only an experiment varying such a quantity discriminates, since one side must lose. A column is *unoccupied* when no coded theory states a prediction over it; an experiment varying it can adjudicate between none of the silent theories, because silence is not a contrary prediction, and a positive result corroborates only whichever account did predict it. Between the two lies the *single-occupant* column, over which exactly one theory or one family states a prediction. A positive result there confirms that account and leaves the others where they were; the informative outcome is the null, which disconfirms the one theory that spoke. A quantity that discriminates must therefore be contested, not merely unoccupied, and must in addition be one that current paradigms do not vary and that is measurable with off-the-shelf methods. Which of the eight quantities coded below falls into which class is the question Section 7 answers, and Section 9 draws the design consequences.

Locating such a quantity requires knowing what the theories actually predict, as opposed to what they are said to predict in review-article summaries; that is this review's contribution. We carried out a purposive audit. The stated predictions of a set of trial-level theories of conscious access, with two origin-level accounts reported apart, were coded against eight measurable quantities from primary statements and the theories' own self-profiling reviews, and the content whose access was manipulated in 36 studies these theories are tested by or built on was classified separately. The audit is not a systematic review, and it reports counts of coded accounts rather than estimates of a population. It is evidence for a claim that is otherwise easy to assert and hard to check: that the field's stated predictions are concentrated on one column of a much wider space, that most of the

remaining columns are unoccupied or single-occupant rather than contested, and that the one contested column cannot yet be measured without report.

Three conventions govern what follows. This is an evaluative review, not a census: every section closes with a judgement about which position the evidence supports, and where a widely repeated claim is weaker than its reputation we say so. We distinguish throughout between claims about *access* — content becoming available to many consumers, and claims about *phenomenal experience*. We use two definitions of “conscious access” and keep them apart, because one of them is not neutral with respect to our own conclusion. The *field definition* is the one in general use: content is accessed when it is reportable and available to multiple downstream cognitive systems. The *working definition* we prefer adds a second clause — that access measurably alters organism state, whether or not it produces fronto-parietal ignition. The second clause builds M5 into the dependent variable, so a reader may reasonably object that the *thinness* of column M5 is partly definitional. It is not, and the coding shows why: each M5 cell asks whether *that theory* states a prediction over organism-wide cortical-autonomic coherence, which is a question about its predictive apparatus and not about our definition of access. Among the coded trial-level accounts, three are coded IMPLICIT there because organism-wide integration is in their spirit but is operationalised cortically, through a single visceral channel, or not at all, and six NO because they locate access in cortical or prefrontal machinery by construction. Neither coding would change under the field definition. What the working definition does affect is how much a thin column *matters*, and that is an argument we make openly rather than smuggle through a definition; Section 7.4 also states why the heartbeat-evoked-response literature, which does couple a cortical measure to a visceral signal, is coded as compatible with M5 rather than as a stated prediction over it. Where a formulation carrying neither commitment is needed we use *global availability*, in the descriptive sense of content being available to many consumers rather than in the theoretical sense the term has inside global workspace theory [4].

The review proceeds from conceptual bookkeeping to measurement. Section 2 asks which distinctions the field has earned. Sections 3 to 5 evaluate the three literatures supplying the raw material for any account of why an organism would need conscious access at all: biological self-maintenance and basal cognition, action arbitration, and valence as organism-wide reconfiguration. Section 6 maps the disputes between theories and what the adversarial test removed from the table; Section 7 presents the audit and classifies each coded quantity as shared, single-occupant, contested or unoccupied; Section 8 examines three unmeasured dissociations; Section 9 states the minimal measurement that follows from the classification. A theory making predictions about the single-occupant coordinates exists and is presented separately; we name it once, at the end, and do not argue for it here.

2. Earned and unearned distinctions

Any claim about consciousness is a claim about one of at least seven things, and most disputes here are about which one. The coding in Section 7 depends on holding them apart: unconscious processing; conscious content; access; online metarepresentation of one’s own processing; control; post-hoc reconstruction; the putative bearer of phenomenal states. These are not degrees of one variable but dissociate pairwise.

The oldest distinction is also the least stable. Block's separation of phenomenal from access consciousness [5] made the inference from "globally available" to "felt" a step requiring argument, later given empirical teeth by the overflow claim that experience contains more than the system can report [6,7]. The programme that grew from it — isolating the correlates of conscious content from its prerequisites and consequences [8], and distinguishing conscious state from content [9] — is a genuine achievement, and the no-report literature of Section 6 its most valuable product.

What has changed is confidence in the two-kinds framing. An argument from inside the experimental camp holds that dividing consciousness into two *kinds* converts a graded multi-component phenomenon into two entities, one inaccessible to measurement by construction [10]. That is a conceptual paper, not a result, but it converges with an independent movement toward feature profiles: multidimensional assessment of animal sentience replacing a single scale [11], indicator lists for invertebrates [12], a framework treating consciousness as a profile of separately assessable capacities [13].

The field has earned the process/content/access distinctions and has *not* earned the phenomenal/access dichotomy in its strong form, and the two are usually run together to the detriment of both. The first set is earned because each distinction is independently measurable and has produced dissociations that survived attempts to explain them away: integration without awareness [14], frontal correlates tracking report rather than the percept [15], conscious state varying independently of content [9]. The dichotomy is not, because its central claim — a second kind of consciousness whose presence comes apart from access — is what no measurement can establish; overflow arguments turn on the interpretation of partial report, the opposing position denies the separation is coherent [16], an intermediate view treats access as partial and hierarchical rather than all-or-none [17].

The cost of conflating these levels shows up in how theories get evaluated. A theory specifying when content becomes globally available answers an access question, and its success says nothing directly about phenomenality; but with the dichotomy in the background such theories are read as accounts of phenomenal experience, criticised for failing a task they never took on. The converse error is as common: theories staking a claim on phenomenality are credited with the access evidence. Feature profiles help by forcing each claim to name its own observable, which is what makes the coding of Section 7 possible.

3. From regulation to a minimal bearer

The literature on biological self-maintenance rigorously accounts for what makes a boundary an organism's own; what it does not give, and largely does not claim to, is any route from organisation to a felt aspect.

Autopoietic theory describes a living system as a network of processes producing its own boundary, treating this closure as sufficient for "cognition" [18]. Di Paolo's refinement is decisive: bare autopoiesis is insufficient even for agency, since a system that either maintains its organisation or disintegrates has no gradations of better and worse and so no norms; what is needed is adaptivity, regulating one's own relation to viability boundaries [19]. **Neither text posits a felt aspect.**

The corresponding corpus is basal cognition [20]. A parallel line proposes the skin-brain thesis: early nervous systems arose not as input–output machinery but to coordinate whole-body contraction [21]. That places single-body coordination at the origin of nervous systems — Section 4’s starting point.

The split is over whether regulation entails experience. A moderate position keeps the functional vocabulary: cellular and bioelectric networks as problem-solvers in morphospace, the self’s computational boundary set by the scale of coordination [22]. A radical position takes the step the moderate one declines, deriving sentience from cellular organisation and locating the first minds in single cells [23] — a redefinition rather than a proposal, adding no mechanism.

Three ladders locate the threshold differently. Feinberg and Mallatt place it in the Cambrian, tied to isomorphic neural representations [24]. Ginsburg and Jablonka propose unlimited associative learning as a *marker*: a detectable sign not a definition [25]. The third is homeostatic: feeling derived from regulation of the internal milieu [26], affect localised subcortically [27], a free-energy formalisation [28].

The anti-cortico-centric line pushes the shared claim — the threshold is not the presence or absence of a cortex — lower than most expect. Merker places the core of the conscious state in upper brainstem and mesodiencephalon, where body schema, action space and motivational ranking integrate, cortex supplying content rather than consciousness [29]. This consolidated into the New York Declaration on Animal Consciousness [30].

The transition from organisation to experience is warranted nowhere here and claimed seriously nowhere except in the radical cellular position — a strength rather than an embarrassment, since the basal cognition programme is valuable precisely because it stops short. Where the inference is warranted is narrower: from organisation to a *control problem* whose solution needs something the organisation does not contain. Coordinating many specialised subsystems around one body is such a problem, solved by nervous systems from their origin [21], and it is where the ladders converge. That convergence is the difficulty: because three independent programmes reach it, “integration around a single body matters” is not an available contribution. The unoccupied position is narrower — the *number and heterogeneity of conflicting domains* integrated is a measurable variable — and no ladder contains that axis, though transitions in cognitive evolution are described as changes in control regime [31].

4. Arbitration as a function

If conscious access exists because incompatible demands from many specialised systems must be reduced to one action of one body, the mechanisms performing that reduction are where to look. Single-body arbitration has been solved unconsciously three times over, in three formalisms, none requiring access.

The first solution is anatomical. The basal ganglia were proposed as the vertebrate solution to the selection problem: many command systems compete for one body’s skeletal motor apparatus, and a centralised device is needed to resolve the conflict without being one of the competitors [32]. Formalisation showed striatal, subthalamic and pallidal microanatomy implementing an off-centre on-surround scheme [33]. This arbiter operates entirely without access [34].

The second solution is computational. Bounded accumulation of evidence to a threshold reproduces reaction-time distributions and accuracy jointly, parietal activity tracking the integral of evidence for an option [35].

The third solution is architectural. The affordance competition hypothesis holds that the visual system specifies several available actions at once while fronto-parietal circuits maintain them as mutually inhibiting population representations biased by value [36]. With two permissible movement directions, monkey dorsal premotor cortex carries two directional activity peaks simultaneously, and only after a cue is one suppressed and the other amplified to threshold [37].

Supramodular interaction theory (SIT) holds that phenomenal states exist to permit interaction among supramodular response systems — agentic, encapsulated structures defined by concerns rather than input modality — which compete for the same skeletomotor output and which, without phenomenal states, could not influence action *jointly* [38]. The passive frame formulation narrows the claim to the somatic nervous system, the mechanism passive and low-level [39].

SIT's empirical base is of three unequal kinds, stronger than its critics allow. The first is taxonomic: introspectively penetrable conflicts (pain for gain, breath holding, delay of gratification) are separated from impenetrable ones (pupillary reflex, ventriloquism and McGurk effects), the dividing line running through a skeletomotor plan. The second is experimental: participants trained to introspect rated the urge to err while sustaining compatible versus incompatible skeletomotor intentions, incompatible intentions producing stronger subjective effects while a smooth-muscle control produced none [40]. That control carries the load: without it the theory would predict a subjective effect from any antagonism. The third support is neuroimaging and markedly weaker: subjective conflict was associated with left postcentral, that is somatosensory rather than executive, cortex [41]. The sharpest question is Clark's: did voluntary skeletomotor action arise for control of the external environment or the internal one, with voluntary regulation of internal-state conflicts as a source of visceral awareness [42].

Our judgement has two parts. First, the three unconscious solutions establish decisively that policy competition is *insufficient* for access: competition is present in all three and access is absent. They establish nothing about its necessity. Necessity would require showing that access does not arise without policy competition, and nothing reviewed here bears on that; §9 accordingly treats it as an untested discriminating prediction. Any theory claiming that access exists *for* arbitration therefore owes an account of what distinguishes the arbitration requiring access from the arbitration that does not; pointing at the existence of conflict is not that account.

Second, SIT does supply such an account, and it is *taxonomic* rather than computational: the conflict either sits at the action-selection stage and involves skeletal musculature, or it does not. This is the strongest available answer, and it pays a price: the classifying feature says where the conflict is located but not what makes it unresolvable without access, so the explanation cannot transfer without re-running the taxonomy. A related cost attaches to the passive frame formulation: the conscious field presents contents but does not select, and operates whether or not conflict is present [43]. But a frame passive with respect to selection has no variable that can differ between arbitration problems, as a discriminating prediction requires. SIT buys defensibility with a commitment hard to

extend, and the cell where an extension would be tested — non-skeletal conflict — is where it makes a clear negative prediction.

5. Valence and organism-wide reconfiguration

An arbitration account needs a common scale. The argument that incommensurable motives must become a single quantity for behaviour to be selectable, with pleasure as that currency, was made decades ago [44], and neuroeconomics gave it a neural reading as a common currency of subjective value [45]. The problem is posed *independently of theories of consciousness*, so a theory of access saying something about the scale makes a claim none of its competitors makes.

The interoceptive literature has meanwhile moved the body from monitored object to control loop, from afferent representation of homeostatic state in insula [46]. The decisive step was redescribing it as *inference* rather than readout [47], regulation proceeding by autonomic, endocrine and behavioural action bringing the body to the predicted state [48].

Organism-wide scope comes from allostasis rather than interoception as such: set points change in advance on anticipated need [49]. Functional connectivity identified a large-scale network named allostatic-interoceptive [50]. Two caveats are load-bearing: that network came from resting-state and meta-analytic methods, not causal manipulation, so it is architectural plausibility rather than demonstrated mechanism; and a recent treatment separates interoception as *modelling* the body from allostasis as *controlling* it [51].

The thesis that an emotion brings many subsystems into a compatible regime is well supported and already owned three times over. Component process theory defines emotion as synchronisation of organismic subsystems [52]. The evolutionary tradition describes emotions as operating modes tuning many parameters for a situation type [53]. Emotion psychology adds control precedence: emotion seizes control and reconfigures action readiness rather than adding content [54]; the constructionist line says the same in allostatic vocabulary [55].

The empirical base is weaker than the theoretical statements suggest. Neuroimaging meta-analysis found no category-specific brain signatures for discrete emotions [56]; meta-analysis of autonomic patterns found the same for the periphery [57]. Global reconfiguration therefore *occurs* but takes no reproducible configuration per category, so the useful unit is a population of states sharing a functional task [58]. Formal models deriving valence from regulatory dynamics rather than positing it — the rate of change of prediction error [59] — are proposals rather than results, but the closest formal analogue available.

Two refinements are non-optional. First, valence is not one parameter: hedonic “liking” and motivational “wanting” dissociate neurochemically and anatomically [60], so a one-dimensional good/bad axis coarsens what is measured. Second, controlling and experienced valence must be separated. The two-system framework holds that subcortical survival circuits produce defensive responses and bodily mobilisation while experienced fear is a separate cortical product [61]; in the stronger version, experienced emotion requires higher-order representation of the state as one’s own [62], resting on the priority of metarepresentation in conscious access [63].

The LeDoux and Brown critique is correct as a constraint and overextended as a conclusion. It is correct that an integrated valent configuration can exist without being experienced, so no account may treat organism-wide valent integration as sufficient for a putative bearer of phenomenal states. It is overextended in inferring that experienced valence requires the higher-order architecture proposed: the step from “integration is not sufficient” to “metarepresentation is what is missing” is not supported by the dissociation data — the affect-first side supplies a rival subcortical mechanism [27,28]. The pain literature is the best model for keeping three levels apart. Nociception is phylogenetically broad and uncontroversial [64]; operational criteria for sentience rest not on it but on features requiring integration — motivational priority shifts, trade-offs, self-administered analgesia [65].

The judgement is that the middle level is real, measurable and under-theorised. Global defensive or appetitive reconfiguration is measurable through motivational priority and trade-off behaviour rather than report, and is where the arbitration and valence literatures meet. What no one has done is make the *structure* of that configuration a function of the *structure* of the conflict resolved: allostatic and constructionist accounts derive affect from regulation and prediction error, neuroeconomics assumes the scale, SIT has no affective variable. The proposition that the number and heterogeneity of conflicting domains sets the required dimensionality of valent integration appears unclaimed — and unlike most such propositions, it is measurable.

6. Theories and their empirical tests

The theories are easier to describe as a list of disagreements than of schools. Almost everyone agrees that conscious content is connected to broad availability; the disagreements are about what makes it conscious — network topology, causal structure, second-order representation, a predictive model of the body. We organise the map by dispute, where the empirical blows of the last five years landed, most striking several theories at once.

6.1 Where content lives

The localisation dispute — posterior versus prefrontal — was the most productive, because both sides stated mutually exclusive predictions. The workspace line ties access to late global ignition in a network privileging prefrontal cortex [66,67], its strongest evidence nonlinearity: a second attentional-blink target’s fate is bimodal rather than graded [68], and the masking threshold coincides with a sharp inflection in late components [69]. Notably, GNWT’s rationale for broadcast is flexibility rather than arbitration — an answer to the frame problem [70], a serial router for composite mental programmes [71]. IIT proceeds from phenomenological axioms to maximally irreducible integrated information, placing the hot zone in posterior temporo-parieto-occipital cortex [72,73], experience identified with the distinctions and relations unfolded from a substrate [74]. Recurrent processing theory holds instead that local recurrent activity in sensory cortex makes content conscious, entailing phenomenal content without access [75,76]. The dispute was long reproduced positionally [77], with the well-founded suspicion that outcomes depend on method and on who runs the experiment [78].

6.2 Who predicts phenomenality

Few theories predict phenomenality rather than access, and the price is metaphysical load or denial of the explanandum. The claim concerns predictive content, not the number of frameworks that address phenomenality. Kuhn's [79] landscape arrays theories of consciousness by ontological commitment — materialist positions in several subcategories, non-reductive physicalism, quantum theories, integrated information, panpsychisms, monisms, dualisms and idealisms — each self-described by its adherents, with critique kept to clarification and no attempt to adjudicate among them; its organising axis is what a theory says consciousness *is*, not which observable it stakes. The inclusion rule of Section 7 cuts across that landscape: a framework enters Table 1 only if it states a prediction over a measurable quantity, so whole regions of Kuhn's map contribute no row — not because they are mistaken but because they answer a different question. Poznanski [80], for example, proposes a constitutive account on which conscious actuality is the intrinsic realisation of a transscale ontophysical continuity; the framework addresses phenomenality at the ontological level and, as presented, states no measurable prediction, and so falls outside the inclusion rule of Section 7. Higher-order theories predict structurally: being conscious is constituted by a second-order representation, entailing dissociations of metacognitive sensitivity with first-order processing intact [63,81]; the most operationalised variant nests a one-dimensional awareness state above a high-dimensional perceptual space [82], with consequences for confusing perception and imagination [83]. Attention schema theory is more radical: a system attributes awareness to itself because it models its own attention [84,85], with an attempt to reconcile AST, GNW, HOT and illusionism [86]. Predictive processing explains not the presence of experience but its structure — perception as controlled hypothesis [87], interoceptive inference as the basis of emotion and bodily selfhood [47,88] — claiming only a basis for correlate-hunting [89]. The poles bound the space: the hard problem as non-derivability of experience from function [90], the meta-problem with illusionism as its answer [91]; the working stake is more modest, making the hard problem operationally easier through accumulated structural correspondences [92].

The methodological threat is logical rather than empirical. The unfolding argument shows that if consciousness is defined by causal structure, a functionally equivalent unfolded network without recurrence behaves identically, including in all reports about experience, while receiving a different value of the measure — so the theory is either false or indistinguishable from its opposite by any behavioural experiment [93]. Responses divided between rejecting its behaviourism [94] and noting a hidden premise about brain states [95]. The problem is deeper than any one theory: falsification requires independent access to phenomenality, and without that bridging assumption every theory is protected from refutation [96] — hence criteria for empirical theories [97], a Lakatosian accounting scheme [98], separating a testable weak core from a loaded strong one [99]. The 2023–2025 controversy over whether IIT is pseudoscientific continues the same dispute, and we report it without taking a side [100,101].

6.3 Access without report

Dependence on report biases toward frontal theories, and the key result is direct: in binocular rivalry, frontal activity tracks introspection and the act of reporting rather than the percept [15]. The programmatic statement [102] and the isolation of perceptual from report-related correlates [103] extend to a report-independent bifurcation signature [104], large-scale networks in a no-report visual

paradigm [105] and audition [106]. Calibration is needed both ways: report reliably adds frontal correlates, but null frontal results do not themselves bury prefrontal theories [107], and no-report designs have their own defect — without report one does not know what the participant experienced [108].

6.4 What Cogitate removed

Exactness matters, because the loose version circulates widely. For IIT the most direct challenge was the absence of sustained within-posterior synchronisation; its content-maintenance prediction passed the pre-set criteria — sustained responses reflecting stimulus duration appeared in occipital and lateral temporal cortex — but sustained representation of orientation did not, although orientation decoded reliably in all three modalities. For GNWT the principal failure was the absence of ignition at stimulus offset, with a secondary challenge on content: category decoded in prefrontal cortex independently of task, identity not at all, orientation only in MEG — raising the question whether prefrontal cortex broadcasts all conscious content or only an abstract part [2].

The adversarial test succeeded methodologically and failed theoretically, and that combination is the most important fact currently available about the field. It succeeded where it matters: predictions fixed in advance, criteria agreed by opponents, results reported against them rather than reinterpreted afterwards. It failed in that the outcome was not fewer live theories but less confidence in a shared dependent variable. The structural lesson is the one the consortium states and the field has not absorbed: a prediction shared with competitors is no asset, because the experiment testing it tests everyone at once. What a new theory can most usefully offer is a prediction about a quantity current designs hold constant — and whether such a quantity exists is answerable by coding the theories.

7. The measurement gap

7.1 What is coded and why

Theories of consciousness are compared, almost exclusively, on where and when conscious contents are encoded. Section 6 traced the consequence: the field's first genuinely adversarial test could challenge both targets while adjudicating neither, and the consortium notes that the challenges extend to other theories sharing the predictions tested [2]. To establish how general this is rather than assert it, we carried out a purposive audit: the stated predictions of a set of theories were coded against eight measurable quantities from primary statements and self-profiling reviews (Table 1), and the content whose access was manipulated in 36 studies these theories are tested by or built on was classified (Table 2). The word *purposive* is doing work. The audit is not a systematic review; there was no registered search, no PRISMA flow and, at the time of writing, one coder (Section 7.2). Its tallies are counts of coded accounts and are reported as such, never as fractions of a population of theories.

The theory set was assembled under one inclusion rule and one coverage aim. The rule: an account qualifies if it has at least one preregistered or explicitly stated empirical prediction about conscious access in the current literature, which excludes purely metaphysical and purely descriptive frameworks. The aim: to span the theory families against which the field's empirical corpus is organised — the four leading theories of the ConTraSt database [78], GNWT, IIT, recurrent processing theory and higher-order theory, together with the attention schema theory — and, in

addition, every family we could identify that states a prediction over one of the five quantities M1–M5 that the corpus does not track: the supramodular family (supramodular interaction theory and its passive-frame successor), the interoceptive-inference family (predictive processing and the beast-machine account, and the neural subjective frame programme), and the two origin-level accounts, unlimited associative learning and the Feinberg–Mallatt scheme. The rule does not pick out these rows and no others. Dendritic integration theory, which predicts that conscious perception depends on coupling between apical and basal compartments of layer-5 pyramidal neurons [109], a prediction tested under anaesthesia [110]; the temporo-spatial theory of consciousness, which states neurophenomenal hypotheses about pre-stimulus temporo-spatial alignment [111,112]; and active inference treated as an account distinct from predictive processing, which has been given a formal workspace model with behavioural and neural predictions [113], all meet the rule and were not coded. We list them so that the omission is visible and so that a second coder can add them; we do not predict here which columns they would occupy, since that is a matter for coding rather than assertion.

Two rows of the earlier version of this table are now reported apart. Unlimited associative learning [114,115] and the Feinberg–Mallatt scheme [24] answer when, and in which lineages, consciousness appeared; they are not accounts of what makes a particular content accessible on a particular trial. Coding them silent on the cortical signature of access records a difference of subject matter, not a hole in a predictive apparatus, and including them in the same tallies as the trial-level accounts inflates exactly the unevenness this review argues for. They therefore form their own block in Table 1, with their own tally, and are excluded from the trial-level counts used in the text. Two rows that were previously merged are now separated. Supramodular interaction theory and passive frame theory share an author and a programme, but Section 4 argues that the passive-frame formulation gives up the variable a discriminating prediction would need; if the difference matters there it must be visible here, and the two are coded from their own statements. Higher-order thought theory and higher-order state space theory remain merged, and we acknowledge that the same issue arises in milder form: the state space formulation adds a stated prediction over metarepresentational precision that the classical formulation does not carry, and a second coder may prefer two rows.

The eight quantities were not chosen for neutrality, and claiming otherwise would be indefensible: three of them (M1, M2, M5) are coordinates of the theory named at the end of this paper. The selection principle was different and is stated here so it can be argued with. Each quantity is a variable that the literatures reviewed in §§3–5 already treat as central to biological regulation, action arbitration and allostasis — not a variable invented for this survey. The test is then whether any theory of *consciousness* predicts over variables that the surrounding biology treats as fundamental. That a coordinate is also an AFT predictor is a consequence of AFT having been built from those same literatures, not evidence that the column set was rigged; but a column set chosen by the author cannot certify its own independence, and the emptiness of a column relative to it is not evidence of anything until the column set has been checked against one the author did not choose. Section 7.5 reports the first such check. The quantities: three concern the arbitration situation (M1 number of simultaneously competing action policies, M2 their degree of incompatibility, M5 organism-wide coherence across cortical and autonomic channels), two the content (M3 valence, M4 autonomic/

interoceptive channel involvement), and three the format and signature of access (M6 dimensionality of phenomenal space, M7 precision of metarepresentation, M8 cortical signature).

7.2 Coding procedure

Each cell was coded YES (a prediction over this quantity is stated and named), IMPLICIT (the theory is compatible with the quantity mattering but formulates no prediction over it), or NO (the quantity is absent from its predictive apparatus). A fourth code, YES (negative), marks a stated prediction that the quantity does *not* affect access. It is a prediction and counts as one, but it is reported apart from YES in Table 1 and throughout the text, because a reader who sees “two theories predict M4” will take the two to agree that the quantity matters, and in this table one of them predicts that it does not. The IMPLICIT/NO distinction carries the analytic weight and we applied it conservatively — a theory was coded IMPLICIT wherever a prediction could be constructed from its stated commitments without adding new machinery, so the count of NO cells is a lower bound on the field’s silence. The boundary is operational: a cell is IMPLICIT when the theory’s primary statement names the quantity, or a quantity it entails, as relevant to access without stating a direction or a measure; it is NO when the quantity does not appear in the statement and cannot be derived from it without an auxiliary assumption the theory has not made. Each non-NO cell is anchored to a specific source; the full matrix with per-cell justifications is in supplementary material.

The first coding was carried out by the first author. A blind second coding is being carried out by the second author using the deposited codebook, without access to the first author’s codes or justifications, on all cells of Table S1 and all rows of Table S2; disagreements are adjudicated by discussion and the pre-adjudication agreement is reported. [$\kappa = \dots$; agreement ... of 88 cells; ... disagreements adjudicated by discussion; S2 $\kappa = \dots$ — to be completed when the second coding is finished] Tables S1–S3, the codebook and the inclusion criteria are deposited at [repository] under DOI [10.xxxx/PLACEHOLDER] and accompany the submission.

7.3 Table 1: theories by measurable quantity

Table 1. Stated predictions of eleven coded accounts against eight measurable quantities. Trial-level accounts of conscious access are tallied first; the two origin-level accounts form a separate block and are excluded from the trial-level tally. Y = prediction stated and named; Y(neg) = stated prediction that the quantity does not affect access; *i* = implicit (compatible, no prediction formulated); · = absent from the predictive apparatus. Condensed from the full matrix (88 coded cells with per-cell citations) in supplementary material.

Theory	M1 N of policies	M2 Incom-pat.	M3 Valence	M4 Auto-nomic	M5 Org-wide coh.	M6 Dim. of phen. space	M7 Metarep. precision	M8 Cortical signature
GNWT	·	·	<i>i</i>	·	<i>i</i>	·	<i>i</i>	Y
IIT (3.0/4.0)	·	·	·	·	·	Y	·	Y
Recurrent processing (RPT)	·	·	·	·	·	·	·	Y
HOT / HOSS	·	·	<i>i</i>	<i>i</i>	·	Y	Y	Y
AST	·	·	·	·	·	<i>i</i>	<i>i</i>	Y

Theory	M1 N of policies	M2 Incompat.	M3 Valence	M4 Auto-nomic	M5 Org-wide coh.	M6 Dim. of phen. space	M7 Metarep. precision	M8 Cortical signature
Predictive processing / beast machine	.	<i>i</i>	Y	Y	<i>i</i>	<i>i</i>	Y	<i>i</i>
Supramodular interaction theory (SIT)	Y	Y	<i>i</i>	Y(neg)	.	.	.	<i>i</i>
Passive frame theory	<i>i</i>	<i>i</i>	<i>i</i>	Y(neg)	.	.	.	.
Neural subjective frame	.	.	<i>i</i>	Y	<i>i</i>	<i>i</i>	.	<i>i</i>
Tally, trial-level accounts (n = 9)	1Y/1i/7.	1Y/2i/6.	1Y/5i/3.	2Y/2Y(neg)/1i/4.	0Y/3i/6.	2Y/3i/4.	2Y/2i/5.	5Y/3i/1.
Origin-level accounts								
UAL	.	.	Y	<i>i</i>	<i>i</i>	.	.	.
Feinberg & Mallatt	.	.	Y	Y	<i>i</i>	.	.	<i>i</i>
Tally, all rows (n = 11)	1Y/1i/9.	1Y/2i/8.	3Y/5i/3.	3Y/2Y(neg)/2i/4.	0Y/5i/6.	2Y/3i/6.	2Y/2i/7.	5Y/4i/2.

Every count below is of the form k of n coded accounts and was recomputed by script from the deposited matrix. Across the 88 cells the codes are 17 YES, 2 YES (negative), 24 IMPLICIT and 45 NO. Per row, counting negative predictions as stated, higher-order theory, predictive processing and supramodular interaction theory state three each; IIT and the Feinberg–Mallatt scheme two; GNWT, recurrent processing theory, the attention schema theory, passive frame theory, the neural subjective frame and unlimited associative learning one each. The spread across rows is modest; the spread across columns is not, and the columns are the result.

7.4 Reading the columns

The columns fall into four classes, and Section 1 explained why the classes matter: what an experiment on a column can show depends on who has spoken over it.

Shared and convergent: M8. The cortical signature of access is predicted by 5 of 9 trial-level accounts and is compatible with 3 more, running from fronto-parietal ignition and P3b for GNWT [67] to early local recurrence for recurrent processing theory [75]. The five predictions share the dependent variable and differ on its locus and timing. That is the column Cogitate tested, and it is why one design could challenge two theories and, by the consortium’s own account, bear on others: every occupant of M8 is at risk from the same measurement, and none can be singled out by it. Dimensionality of phenomenal space (M6) and precision of metarepresentation (M7) are each predicted by two theories, and sharply, by higher-order state space theory [82] and by IIT [74]; both columns are occupied but not contested, since the two predictions in each concern different aspects of the quantity and no coded theory denies either.

Single-occupant: M1 and M2. The number of simultaneously competing policies (M1) and their degree of incompatibility (M2) are each predicted by 1 of 9 trial-level accounts, supramodular interaction theory, whose PRISM principle makes the arrival of multiple simultaneous inclinations at the skeletomotor output system the trigger for phenomenal involvement [38]. Incompatibility there has been measured — sustaining incompatible skeletomotor intentions while motionless produces stronger subjective effects than sustaining compatible ones [40]. Passive frame theory, coded separately, is IMPLICIT on both: the function of the conscious field is still stated as the integration of skeletomotor inclinations, but the field is said to present its contents whether or not conflict is present [39,43], so neither the number of inclinations nor their incompatibility is a predictor of access on a trial. Predictive processing is IMPLICIT on M2, and no other coded account states a prediction over either quantity. These are single-occupant columns in the sense of Section 1. An experiment varying incompatibility cannot adjudicate between GNWT, IIT, recurrent processing theory or the attention schema theory, none of which said anything about it; a positive result corroborates the supramodular prediction and leaves the rest untouched, and the informative outcome is the null.

Contested: M4. Autonomic involvement is the one column carrying opposed stated predictions. Among trial-level accounts it has 2 YES, 2 YES (negative) and 1 IMPLICIT. Predictive-processing accounts make visceral channels constitutive of emotion and embodied selfhood [48], the neural subjective frame makes cortical monitoring of cardiac and gastric input the generator of the first-person perspective within which contents become conscious [116], and the affective branch of the Feinberg and Mallatt scheme, in the origin-level block, agrees [24]. Against them, supramodular interaction theory predicts testably that smooth-muscle conflict produces no comparable change in experience [40], and passive frame theory states the same boundary as a claim about scope: the conscious field serves the somatic nervous system alone [39]. This is the cell an adversarial test should want. It cannot yet be run. Report-independent access measures are established in the visual [15,103] and auditory [106] modalities, and their logic — dissociating a stimulus-locked neural marker from the report that usually accompanies it — has no validated interoceptive counterpart; cardiac interoceptive accuracy measures require report by construction [117]. This is a methodological gap rather than a theoretical one, and its consequence is that the condition of Section 9 that would discriminate needs a validation study before it can be run, not merely a protocol.

Thin, not empty: M5. Organism-wide coherence across cortical and autonomic channels, a joint measure of the state accompanying access, receives 0 YES and 3 IMPLICIT among 9 trial-level accounts (0 YES, 5 IMPLICIT among all 11 rows). Three trial-level accounts are compatible with it and none operationalises it as an organism-wide joint measure. GNWT's global availability is organism-wide in rhetoric and cortical in measurement, its dependent variables being ignition and P3b [118]; allostatic accounts identify a large-scale interoceptive system [50] without proposing its coherence with cortical dynamics as an access criterion. The nearest stated prediction is the third IMPLICIT cell, and it deserves to be described rather than filed. The neural subjective frame proposal holds that cortical responses to heartbeats are a precondition of conscious perception [116], and pre-stimulus heartbeat-evoked responses do predict visual detection [119] and somatosensory detection [120,121], as does respiratory phase [122,123]; heartbeat-evoked responses have also been used to detect residual consciousness in post-comatose patients [124]. These measures couple cortex to one

visceral channel and are, by the original authors' own control analyses, independent of the measured autonomic state: in the founding study neither fluctuations in bodily parameters nor overall cortical excitability accounted for the effect [119]. They therefore index cortical monitoring of a visceral signal rather than coherence across cortical and autonomic channels, and the programme itself frames the quantity as necessary but not sufficient for conscious perception [116] and as distinct from attention [125]. We code it IMPLICIT on M5 and YES on M4, and we withdraw the earlier formulation that nobody operationalises a cortico-visceral quantity as a condition on access: somebody does, single-channel, and the column is thin rather than empty. A coder who read the heartbeat-evoked response as a cortical–autonomic coherence metric would fill 1 of 9 trial-level cells, and Section 7.7 carries that sensitivity.

Occupied without opposition: M3. Valence is predicted by 1 of 9 trial-level accounts (predictive processing) and by both origin-level accounts, and is IMPLICIT for 5 more; no coded theory denies that valence bears on access, so the column is occupied but not contested.

The columns that are not shared intersect in a cell no study fills: access to valenced or interoceptive content under a manipulated number of incompatible policies, measured across cortical and autonomic channels jointly. Part of that cell is occupied, and by whom is worth stating precisely. Morsella and colleagues established that policy incompatibility has measurable consequences for experience and that smooth-muscle conflict does not [40,126]; Desender and colleagues established the more directly relevant point that the subjective experience of conflict is a separable measurable quantity, dissociable from conflict adaptation — adaptation occurs without conflict awareness, and awareness of conflict has effects of its own [127], which is the closest existing precedent for treating M2 as a predictor of experience rather than of performance; Garfinkel and colleagues, that interoceptive accuracy dissociates from interoceptive awareness [117]. What none has done is vary conflict load orthogonally to salience while measuring access with a report-independent signature, the constraint the no-report literature established as necessary [102], with objections in [108]. Nor can existing data be redescribed to fill the cell: no reanalysis of the Cogitate dataset can recover an effect of policy conflict, since its stimuli — faces, objects, letters and false fonts under target detection, hold conflict load constant by construction [2].

A gap reducible to unavailable methods is a difficulty, not a gap. Most of what the missing measurement requires is routine, and one component is not; the component that is not — a report-independent measure of access to interoceptive content — was named above under M4, and it decides which parts of the design in Section 9 are ready to run. Routine: policy conflict is manipulable and its subjective consequences measurable without overt movement [40]; report-independent access measures are established in the visual [15,103] and auditory [106] modalities; calibrated valenced stimuli [128] and standardised cardiac interoceptive measures [117] are off the shelf; and multi-site preregistered adversarial designs are feasible at scale [1]. The absence is explained instead by what the theories ask for. The field's own coding shows that support for a given theory is predictable from methodological choices alone, irrespective of findings, and that most studies interpret results post hoc rather than testing a priori predictions [78]. Where no theory names conflict load or organism-wide coherence as a variable, no paradigm is built to vary it.

7.5 An independently generated column set

Three of the eight columns are the author's own, and the sensitivity check offered in the earlier version of this section — dropping M1, M2 and M5 and confirming that M8 still dominates — tested a proposition nobody was going to contest. The proposition that needs testing is different: whether the emptiness or thinness of a column is a fact about the theories or an artefact of a column set chosen by someone whose theory occupies it. Emptiness relative to a column set of the author's choosing is not evidence of anything until the column set is independent. The first check we could run uses the dimensional scheme of the ConTraSt database [78], assembled by people with no stake in this argument to organise 412 experiments, and codes the same theory rows against its twelve dimensions with the same four codes (Table S3; 120 cells, ten rows, since the neural subjective frame was added after that coding).

The result has two parts. First, along the independent column set every dimension receives at least one stated prediction; none is empty in the sense that M5 is thin in Table 1. Occupancy is uneven in a way that mirrors the corpus: seven of the twelve dimensions — experimental paradigm, measure type, task type, spatial findings, dependent measure, stimulus, and report versus no-report — carry stated predictions from at least four of the eight neuroscientific rows, while the sparse dimensions are the electrophysiological result fields — frequency findings receive one stated prediction (GNWT) and temporal findings three (GNWT, IIT, recurrent processing theory). Per row the ordering also mirrors the corpus: GNWT states predictions on 11 of 12 dimensions, IIT on 10, recurrent processing theory on 7, passive frame theory on none, because it declines the implementation level. The theories are therefore not uniformly underspecified, and the thinness of M5 is not an artefact of general vagueness: the same accounts that are silent on M5 speak on most of the axes the field's own inventory is organised on. Second, none of M1, M2 or M5 corresponds to a ConTraSt field. The check covered every field name and value in the live schema. No field counts concurrent response options; response conflict is not an annotated variable, the conflict paradigms sitting under a categorical task label with no conflict-magnitude field; and no field records cortical–autonomic coherence, although its components exist separately as cortical connectivity tags and cardiac tags, including the heartbeat-evoked potential. By contrast M7 and M8 are directly represented, and M3, M4 and M6 are partly recoverable. The quantities this review calls unmeasured are also unrecorded in the largest theory-neutral inventory of the field.

What this does and does not establish should be stated with equal care. It establishes that the Table 1 result for M1, M2 and M5 is not the rediscovery of an existing axis under another name, and that the accounts coded silent on those columns are not silent in general. It does not establish that the quantities are absent from experiments: ConTraSt records what authors reported and interpreted, and a conflict-magnitude or cortical–autonomic variable could have been manipulated in a coded study without acquiring a field. It cannot say whether M5 is thin because the quantity is hard to measure or because theories find it uninteresting, and it says nothing about whether M1 and M2 are the right variables, only that they are orthogonal to the scheme. The coding has its own limits: the published extraction sheet was inaccessible, so the scheme was reconstructed from the database's methods description and live schema; several GNWT and IIT cells rest on the Cogitate preregistration rather than on foundational texts, because that is where proponents committed to dimension-indexed predictions; the YES code admits explicit invariance claims, which inflates three dimensions, and

under a strictly differential reading GNWT falls to 9 stated predictions, IIT to 8; and it was done by one coder and is included in the second-coder package. The second independent route — writing to proponents of four or five of the theories, asking each which quantities they take their theory to be committed to, and coding against the union — has not been taken, and we name it as the next check rather than claim its result.

7.6 Table 2: content inventory

The inventory asks a different question of the same theories: what content have their tests actually manipulated? The 36 studies are a purposive sample assembled from the primary statements of the coded theories, the studies those statements cite as support, and the adversarial-test literature, under the inclusion criteria deposited with the tables. Content was coded by the content whose *access* was manipulated, in six mutually exclusive classes. The category that does the analytic work is “theory-addressed”, and its operational rule is criterion I2 of the inclusion criteria: a study is theory-addressed when it is cited as evidence for, or presented as an explicit test of, at least one theory in the prediction matrix, or is a canonical paradigm paper on which such tests are built. A study of conscious access that none of the coded theories cites or tests is in the inventory but is not theory-addressed.

Table 2. Content whose access was manipulated in 36 studies that the **coded** theories are tested by or built on. The third column restricts to studies addressed at one of the coded theories, which is the population relevant to the claim about theory-driven design. Full per-study coding with paradigm and theory attribution in supplementary material.

Content class	All 36 studies	Of 33 with a content manipulation	Of 27 theory-addressed with a content manipulation
Neutral visual	22	22	22
Competing motor policies	4	4	4
Valenced	4	4	0
Interoceptive	2	2	0
Neutral non-visual	1	1	1
No content manipulation (state)	3	—	—
Total	36	33	27

Of the 36 studies, 30 are theory-addressed and 6 are not, and the six are exactly the four valenced and the two interoceptive studies: the valenced studies [128–131] and the interoceptive studies [117,132]. The zero in the third column is therefore produced by the theory-addressed filter, and that is the finding, not an artefact to be explained away: the valenced and interoceptive access literature exists, is methodologically sound — fearful expressions break continuous flash suppression faster than neutral ones [128], threat-signalling input gains preferential access [130], arousal and valence have been varied orthogonally [129] — and is claimed as evidence by no coded theory. Among the 27 theory-addressed studies with a content manipulation, 22 used neutral visual material, 4 competing motor policies (all from the supramodular programme) and 1 auditory no-report content [106] as the sole non-visual case.

How sensitive the zero is to the filter is stated in full. As coded, valenced content is 0 of 27 and interoceptive content 0 of 27. Moving one study across the line — counting Gayet et al. [130] or Yang

et al. [128] as addressed at a coded theory, which a reader who regards continuous flash suppression as a GNWT-relevant paradigm might do — gives valenced 1 of 28 and neutral visual 22 of 28. Counting every content study regardless of theory linkage gives valenced 4 of 33, interoceptive 2 of 33 and neutral visual 22 of 33. The zero does not survive every reading; the imbalance does, and it is the imbalance that the abstract carries. The pattern is corroborated from outside our own coding: the field's census of 412 experiments codes stimulus modality and category as methodological variables and reports no valence breakdown at all [78]. One further caveat belongs here. The inventory was built for the original theory set; the studies on which the neural subjective frame row rests manipulate neutral visual or somatosensory content with a visceral predictor [119,120], so adding them would enlarge the neutral cells and leave the valenced and interoceptive cells where they are. The inventory is left as deposited pending the second coding.

7.7 What survives reclassification, and limitations

Because the IMPLICIT/NO boundary carries the weight, we tested the structural findings against the least favourable reclassification available: promoting *every* IMPLICIT cell to YES, that is, deciding every judgement call against our own conclusion. The M1 finding survives: the column has 1 YES and 1 IMPLICIT among 9 trial-level accounts, and the IMPLICIT cell is passive frame theory, so promotion turns one occupant into two from the same family, and no other family speaks on any reading. The M2 finding survives in the sense that matters for Section 9: 1 YES and 2 IMPLICIT become at most 3 of 9, one of them predictive processing, so the column is single-family as coded and at most two-family under promotion; no reading makes it contested, since no account predicts that incompatibility does *not* affect access. The M5 finding does not survive: promoting its three IMPLICIT trial-level cells turns a column with no stated prediction into 3 of 9 (5 of 11 across all rows), and a coder who read the heartbeat-evoked response as a joint cortical–autonomic measure would fill one cell even without promotion. The claim that M5 is thin therefore rests on the judgement that a rhetorical organism-wide commitment without an operationalisation, and a single-channel cortico-visceral predictor, are not stated predictions over organism-wide coherence. We think that judgement is right and have given the per-cell reasons in Section 7.4, but it is a judgement, which is why M5 is a secondary and qualified claim in this review and the single-occupant status of M1 and M2 is the headline. The M4 classification as contested is unaffected by promotion, because it rests on YES and YES (negative) cells, not on IMPLICIT ones. The sensitivity analysis is deposited with the coding.

Four limitations bear on how much weight Tables 1 and 2 can carry, and we state them without mitigation. First, the codes reported here are those of a single coder; the blind second coding by the second author is described in Section 7.2 and its agreement statistics are reported there once complete. Until then the entire evidential base of this section is 88 cells and 36 study classifications produced by one reader applying a written scheme to texts, and we make no reliability claim. Second, both the theory set and the study set are purposive samples. The inclusion rule for theories is stated and the accounts that meet it and were not coded are named in Section 7.1; a registered systematic search for studies would very likely change the counts in Table 2, substantially so for the smaller cells. Third, the column set is the author's, and the only independence check performed is the ConTraSt coding of Section 7.5; the letter-to-proponents route, which would generate a column set from the theories' own advocates, has not been taken. Fourth, coding a “predictive apparatus”

requires deciding which formulation of a theory counts as canonical, and for theories with several revisions we coded the most recent statement making empirical commitments — defensible but not neutral, and the reason supramodular interaction theory and passive frame theory are now two rows rather than one.

8. Online metarepresentation, policy candidacy and reconstruction

The upper part of the episode — registering one’s own processing, treating a proposed action as one option among several, reconstructing it afterwards — holds three dissociations that are framed but, with the partial exception conceded in Section 8.2, not measured, each conflated with a measured neighbour.

8.1 Precision, not registration

The case for treating self-registration as a separate quantity is not introspective: the accuracy of judgements about one’s own processing varies separately from that processing’s accuracy. Second-order signal detection supplies the apparatus, meta- d' expressing metacognitive sensitivity in units of first-order d' [133]; calibration, error sensitivity and performance dissociate, so “metacognition” unqualified means little [134]. Theta-burst TMS to prefrontal cortex impairs metacognitive awareness with discrimination intact [135]; lesion data implicate anterior prefrontal cortex, whose integrity predicts introspective rather than perceptual accuracy [136,137], with domain-general and domain-specific components [138] and a separate evaluative stage [139]. Registration itself grants no control. But its *precision* predicts downstream change — confidence governs information search [140], poor metacognition accompanies resistance to counter-evidence [141], symptom dimensions shift metacognitive bias without touching performance [142]. The measurable variable is metarepresentational precision, not the fact of registration — a different claim from the one most theories in Table 1 make.

Two qualifications. Self-registration may not be self-directed: confidence compares contributions to joint decisions [143], is arguably a culturally acquired skill over implicit monitoring [144], secondary to mind-reading [145], and what a workspace needs to select what is admitted [146]; the radical version makes consciousness a learned metarepresentation [147–149]. And these claims rest on non-equivalent report scales: the Perceptual Awareness Scale exists because binary seen/unseen flattens intermediate states [150] and is more sensitive to weak contents than confidence [151], the discreteness dispute is unresolved [152] against workspace theories predicting a threshold [153], and scale and quantity are poorly coordinated [154].

8.2 Policy candidacy

The second dissociation concerns **policy candidacy** (polyvariant choice) — the mode in which a proposed action policy ceases to be the sole command and becomes one member of a compared set. Three features define it: a change in the proposal’s status, a comparison set, and independence from both online metarepresentation and the success of control.

The name requires justification, because the obvious label is taken. In computational neuroscience *meta-control* means arbitration between controllers and tuning of their parameters: model-based versus model-free learning by reliability [155], deciding how to decide [156], allocating effort by expected benefit [157], reviewed in [158]. That is parametric arbitration, not a policy represented as

an object; since the term denotes the very neighbour this mode must be distinguished from, we do not use it, and this is its only appearance.

Existing paradigms measure everything adjacent to policy candidacy and not it. Meta-d' measures the accuracy of evaluating a decision already taken [133]. The expected value of control measures the *intensity* of allocation in dorsal anterior cingulate cortex [159,160] over a cascaded prefrontal organisation [161] whose capacity limit is probably task separation, not fuel depletion [162] — the resource model having failed a preregistered replication [163] and multi-site test [164], effort recast as opportunity cost [165]. The stop-signal paradigm measures the speed of cancelling one prepared action [166,167]; its unconscious variant is decisive, masked stop signals recruiting frontal inhibitory machinery without awareness [168,169], conscious and unconscious control engaging dissociable rather than graded mechanisms [170]. Conflict detection likewise has a canonical account in which the anterior cingulate recruits control with no commitment to consciousness [171]. The closest construct is intentional inhibition, where refusal is itself a choice [172,173] — but there too no alternative is generated and compared.

A correction here strengthens the case. The first of the three quantities operationalising policy candidacy — the size of the set under consideration — *is* already measured by option generation [174,175], including its neural basis [176], so the claim that none is measurable is too strong. Unmeasured are whether a decision depends on properties of the *rejected* alternatives, and whether metacognitive precision predicts option-set widening rather than correctness. The metric for the first, **option-set widening**, should be borrowed from option generation, not reinvented; the nearest precedent for the second is that the selection threshold shifts with urgency [177].

8.3 Report as write-back into the self-model

The third dissociation concerns what report is evidence for. That participants report not the causes of their behaviour but plausible theories of those causes remains established, if localised more precisely than originally [178], extending to introspection about current phenomenal content [179]. Choice blindness is the strong version: when already-chosen faces were switched, many switches went unnoticed and participants justified a choice they had not made [180], reproduced outside the laboratory [181].

The result that matters most is stronger than the standard reconstruction claim: switching participants' own answers in a moral survey shifted the attitudes themselves, durably [182]. Reconstruction is therefore not a distortion of the record but a write-back into the model of oneself — report can become a cause of the subsequent state. The same holds for the sense of agency, an inference rather than access to the motor command [183,184], and for the construction of having chosen freely [185]. Two cautions: intentional binding [186] is contested, reproduced under illusory voluntary action [187] and sensitive to phenomenological control [188]; and the Libet result must be stated exactly — the averaged readiness potential [189] is not refuted as a measurement, its causal interpretation is, since stochastic accumulation makes it an artefact of averaging [190,191]. Hence “post-hoc reconstruction” rather than “reflection”: the latter imports a transparency the data do not support, itself analysed as the unavailability of the model's own modelhood to the system using it [192–194].

8.4 What is measured and what is not

The three dissociations do not stand in the same place, and the ledger is uneven rather than empty. Metarepresentational precision has validated instruments — meta- d' and its calibration variants — and no experiment that asks whether precision, rather than registration, gates access; that test can be run with what exists. Policy candidacy is measured in part: option generation already counts the set under consideration (Section 8.2), whereas the dependence of a decision on properties of the *rejected* alternatives has no measurement at all. Write-back has been demonstrated [182] but has not been turned into a measure that any theory is committed to. What the three share is therefore not absence of measurement but absence of a claimant: none is the dependent variable any theory in Table 1 predicts, and Section 7 documents what that costs. Each would be cheaper to test than the localisation dispute that has absorbed the field's resources; whether any of them is tested depends on whether some theory is prepared to put a prediction on it.

9. Conclusion: the minimal measurement

The field's problem is not a shortage of theories or of rigour. It is that the coded theories concentrate their stated predictions on one of eight measurable quantities, so a well-run adversarial experiment tests many of them simultaneously and none distinctively. That is what Cogitate demonstrated, and the consortium said so: the challenges extend to theories sharing the predictions tested [2]. Two coding results locate the columns that are not shared, and the classification of Section 7.4 says what each can carry. The number and incompatibility of competing action policies are single-occupant columns: supramodular interaction theory predicts over both and confines the prediction to skeletal muscle by construction [38,39], and no other coded account states a prediction over either. Organism-wide coherence across cortical and autonomic channels is stated by no coded theory and is compatible with three. The one contested column is autonomic channel involvement, where two families have opposed predictions on record. And in the theory-addressed literature the valenced and interoceptive studies are absent because no coded theory claims them, not because they do not exist.

One experiment converts the gap into a result. Hold salience and task relevance fixed. Vary the number of mutually incompatible action policies that the content licenses, across at least three levels. Run it once with neutral and once with valenced content matched on salience. Measure access with a report-independent signature alongside a simultaneous autonomic channel.

The design needs one control that the coding scheme makes obvious and that a naive version would omit. Increasing the number of mutually incompatible policies also increases cognitive load, and load degrades access under every theory in Table 1. GNWT in particular, joined to any capacity assumption, predicts that several competing policies consume workspace resources and so degrade access; that is a prediction, and Table 1 does not code GNWT against it, because it is a prediction about load rather than about the policies' incompatibility. What GNWT and the other coded theories do not predict is an effect of *incompatibility at matched load*. The design therefore holds the number of candidate policies, their salience and the working-memory demand constant and varies only whether the policies are mutually exclusive: the same option set, the same load, incompatibility present or reduced to zero. The supramodular prediction is an effect of incompatibility with policy count fixed; a pure capacity account predicts an effect of count with incompatibility fixed. Only the

crossing of the two licenses a claim about conflict. Three quantities from Table 1 then become jointly observable for the first time, and each of the three outcomes is informative in a different way.

9.1 One worked condition: incompatibility $\times$ valence under no-report

Design and participants. A within-subject 3×3 crossing of the incompatibility of the action policies the content licenses (I1, I2, I3) with content valence (negative, neutral, positive), policy count fixed at three, plus a neutral-only load-control arm varying count (one, two or three compatible policies) at zero incompatibility, the crossing §9 requires. Two sessions of about 2.5 h, at least 48 h apart. Participants: healthy right-handed adults aged 18–40, normal or corrected acuity and colour vision (hue is a rule feature), no neurological, psychiatric or cardiovascular history or related medication; N follows from the power analysis below. Preregistered exclusions: below 90% accuracy on full-contrast rule trials after training; titrated visibility outside 35–65%; cross-validated decoder AUC below 0.70; over 30% of trials lost to artefact; own ratings failing to order the faces negative < neutral < positive.

Stimuli. Targets are grey-scale faces of six identities (three female) in sad, neutral and happy expressions from the Karolinska Directed Emotional Faces set, whose validation supplies hit-rate, intensity and arousal norms [195]. Sad rather than fearful expressions are used because threat gains access through routes that need not involve valence [128,130]. Sad and happy exemplars are matched on normative arousal; neutral exemplars are necessarily lower, so the critical valence contrast is negative against positive at matched arousal, with trial-wise pupil and skin-conductance responses as covariates in every model. The 18 images are equated on luminance histogram and spatial-frequency spectrum with the SHINE toolbox [196], cropped to an oval, shown at 4° . Each face carries three task features orthogonal to expression: isoluminant tint (three hues), position (above, at or below fixation) and aperture shape (circle, square, diamond).

Task and incompatibility manipulation. Participants overlearn three rules — hue $\rightarrow$ key, position $\rightarrow$ key, shape $\rightarrow$ key — over the same three keys. Instructed rules activate their responses on stimulus presentation without being executed [197,198] — operationally, the content licenses a policy. All three rules hold on every trial, so policy count is three throughout; incompatibility is fixed by the target’s feature conjunction: the three demanded keys identical (I1), two identical (I2), or all different (I3). Rule assignment is rotated across participants and every feature value occurs equally often at every level, so stimulus set, salience and task relevance are identical across levels by construction; only the conjunction differs. A trial runs: fixation 500–800 ms; target 33 ms at an individually titrated contrast; pattern mask at 50 ms SOA for 200 ms, staircased to about 50% visibility [69]; 1,000 ms blank. On 25% of trials a probe names one rule and the participant executes it, or presses a fourth key for “no face”; on 75% nothing is required. Probe trials keep the rules active and are the manipulation check, gating the analysis: response time, error rate and lateralised readiness potential must rise monotonically $I1 < I2 < I3$ at full contrast. The load-control arm (neutral faces) has one or two rules in force, targets chosen so the active rules agree, giving with I1 count one, two, three at zero incompatibility. Trials per cell: 40 in the report session (360 trials) and 64 in the no-report session (nine cells plus two load-control cells, 704 trials), intermixed in blocks of 44; about 4.5 s per trial, 6 s with the rating.

Report-independent signature. Session 1 adds a Perceptual Awareness Scale rating after the probe phase on every trial [150]; session 2 repeats the stimulation without it. The access index is trial-wise

decoded evidence for “seen” from a classifier trained on rating-labelled session-1 trials, using posterior EEG in the 100–350 ms window — where the visual awareness negativity survives removal of report while the P3b does not [103,199–201] — with pupil, blink and microsaccade features — eye measurements of the kind shown to classify perceived from unperceived trials without report [105]. Applied to session-2 no-probe trials, it gives the cell-level dependent variable, mean decoded evidence (logit). Two preregistered conditions guard it: trained on one incompatibility level it must generalise to the others (indexing seeing, not conflict), and it must reproduce session 1’s rating-based effects under cross-validation. Its limits: inference from physiology to a conscious state is theoretical, and a no-report participant may still covertly categorise [102,108]; the transfer assumes a seen/unseen signature invariant to report demand, and which windows are invariant is unsettled — the P3b vanishes without report in masking [199] while late sustained activity persists in task-free listening [104]. Every contrast is therefore also computed on the session-1 rating, and a result is claimed only where both arms agree.

Cortical and autonomic recording. 64-channel EEG (0.1–40 Hz; epochs –200 to 800 ms), lead-II ECG, skin conductance and pupil are recorded throughout. The heartbeat-evoked potential (HEP) is the mean fronto-central amplitude 250–400 ms after the last R-peak preceding the target by at least 400 ms, cardiac-field artefact controlled by surrogate-R-peak permutation. Pre-stimulus HEP predicts detection of weak visual [119] and somatosensory stimuli [120] with opposite signs, so its direction is preregistered two-sided; cardiac phase at onset is a covariate. Phasic skin conductance is deconvolved 1–5 s post-target [202,203]; pupil dilation is the baseline-corrected mean 0.5–2 s post-target [204]. The cortical–autonomic coupling index, the operational proxy for M5, is the per-participant, per-cell standardised slope of decoded access on pre-stimulus HEP, and the trial-wise correlation of decoded access with the phasic pupil and skin-conductance responses.

Hypotheses as contrasts, and who has predicted what. C1: linear trend of access over $I1 < I2 < I3$ at fixed count (weights –1, 0, +1). The M2 column of Table 1 holds one YES, supramodular interaction theory, whose evidence is stronger subjective effects when sustaining incompatible rather than compatible skeletomotor intentions [40]; extending this to access is our derivation, for its proponents to endorse or reject before data are collected. Predictive processing is IMPLICIT; the other seven are NO — silent. A positive C1 therefore confirms one theory and refutes none: GNWT, IIT, RPT, HOT and AST are untouched, because silence is not a contrary prediction. It becomes adversarial for GNWT only if its proponents commit in advance to the null that content-neutral broadcast suggests — what an adversarial preregistration exists to solicit [1]. C2: the load-control trend over count at zero incompatibility; a pure load account predicts a negative slope, no theory a positive one, and C1 is attributable to conflict only if C2 is flat or negative. C3: valence, as (negative + positive)/2 – neutral and as negative – positive at matched arousal. On M3 predictive processing is YES, GNWT and HOT IMPLICIT, the evolutionary accounts YES at lineage rather than trial level, the rest NO. A C3 effect fills an unoccupied cell and discriminates only if GNWT’s content neutrality is stated as a null beforehand. C4, the interaction (C1 slope for valenced minus neutral content), and C5, the M5 coupling slope and its change with incompatibility, are unoccupied by all nine theories; the companion theory predicts both positive, and both are preregistered as estimation, not confirmation. The genuinely adversarial cell in Table 1 is M4 — supramodular interaction theory’s stated negative against the interoceptive-inference YES [39,48] — and this condition does not test it. Testing it needs

a report-independent measure of access to interoceptive content: a signature tracking, trial by trial, whether an interoceptive event (a heartbeat, a respiratory load) was accessed, validated against report in a calibration set, generalising to no-report blocks, and dissociable from afferent strength, from arousal and from attention to the body, which alone modulates the HEP [205,206]. None exists; the HEP indexes attention and arousal, not access, and the M4 test waits on its development.

What each outcome would show follows from the classification in Section 7.4, and we state it without inflation. A positive effect of incompatibility at matched load corroborates the supramodular prediction and refutes no silent theory: GNWT, IIT, recurrent processing theory and the attention schema theory said nothing either way about incompatibility, and silence is not a contrary prediction. The informative outcome is the null. At adequate power, the absence of an incompatibility effect with load and salience matched would disconfirm the only stated prediction over M2 and would leave the arbitration reading of access without its one measured predictor. A difference in access threshold between matched valenced and neutral content occupies a cell no coded theory claims, which makes it the cheapest way to convert theory-orphaned evidence into theory-relevant evidence — again without discriminating between the coded theories. The contested cell is M4, and it is the only one in Table 1 where a result would count against a stated prediction whichever way it came out: supramodular interaction theory and passive frame theory predict no access for a purely autonomic–interoceptive conflict without skeletomotor output, while interoceptive-inference accounts predict that there is [39] against [48], and the neural subjective frame makes cortical monitoring of visceral input a precondition of access [116]. The opposition is on record from both sides; what is not on record is a way of measuring the outcome.

Power. Per-participant contrast scores (paired t , $\alpha = 0.05$ two-sided), confirmed by trial-level mixed-model simulation with participant random intercepts and slopes [207,208] (`section9_power.py`, `section9_power.csv`), give 90% power at $N = 68$ for the incompatibility trend at $dz = 0.40$ and $N = 171$ at $dz = 0.25$; for the interaction, $N = 119$ at $dz = 0.30$ and $N = 265$ at $dz = 0.20$. Trials per cell matter little once participant slopes are heterogeneous: in the two-stage simulation (2,000 runs, N in steps of 10) going from 32 to 64 trials changes the N required by 0–10; the coarser mixed-model grid (200 runs per point) shows differences of 0–35 that are within its Monte Carlo and grid resolution. The 64 no-report trials per cell are set by decoder reliability. The planning values sit below the nearest published effects, all of which retained variance this design removes: incompatible versus compatible intentions on report-based ratings, $t(13) = 6.21$ [40]; working-memory load on masked-detection d' , $d = 0.62$ – 0.86 [209]; cardiac phase on detection, $dz \approx 0.65$ [120], with Park et al. [119] as the visual anchor for the HEP–detection link; meta-analytic HEP effects of $g \approx 0.4$ [206]. The confirmatory study is set at $N = 126$ (a multiple of the six rule assignments): 90% power for the trend at $dz \approx 0.29$, for the interaction at $dz = 0.30$, and for equivalence within $dz = 0.30$ by two one-sided tests [210], so a null on C1 is a supported null; the small interaction is under-powered and reported as estimation. Sequential Bayes-factor monitoring runs from $N = 60$ in batches of 18 against an interval null [211], maximum $N = 150$.

The third condition’s design requires care, and we flag the difficulty rather than leave it for a reviewer, because it is where the prediction could be rendered unfalsifiable in either direction. Most homeostatic crises recruit skeletal effectors — respiration through the diaphragm and intercostals,

thermoregulation through shivering and posture, so breath-holding is not evidence against a skeletomotor boundary. What is required is a conflict whose resolution is wholly autonomic: antagonism between thermoregulatory and gastrointestinal demands with posture fixed and respiratory pattern change blocked, or orthostatic and thermal loading of the baroreflex without postural correction. De-efferentation (high spinal injury, locked-in syndrome) is not a discriminator either, since the claim concerns intentions rather than execution and the central skeletomotor plan is preserved. Blocking respiratory pattern change is itself a skeletomotor intervention and does not escape the objection: in an awake participant, skeletomotor intentions survive any fixation of posture or breathing, and supramodular interaction theory is a claim about intentions.

The deeper obstacle is the one named in Section 7.4: the third condition requires a report-independent measure of access to interoceptive content, and no such measure has been validated. This is why we present the third condition as the most discriminating test *in principle* and not as the one to run first. The order of work is the reverse of the order of interest — the crossed conflict and valence design uses established visual and auditory no-report measures and can be run now; the autonomic-only condition needs a validation study developing and calibrating an interoceptive no-report signature before it can adjudicate anything. The situation should be stated as it is: the condition that can be run tests a single-occupant prediction, and the condition that would discriminate between stated predictions cannot yet be run. That is the actual conclusion of this review, and it is a smaller claim than an adversarial test on a new quantity — but it names the measurement whose absence keeps the contested cell closed. Presenting the third condition as ready would be the kind of overclaim this review has been criticising.

A closing note on what this review has and has not done. We have not proposed a theory. We have argued that the field’s measurement effort is concentrated on a quantity shared across theories, documented that concentration by coding, and specified the experiment that would test what the coding shows to be untested. A theory that does make predictions about the single-occupant coordinates — about the format in which incommensurable local evaluations become comparable, its dimensionality as a function of the number and heterogeneity of conflicting domains, and organism-wide reconfiguration breadth as a second condition on access, exists and is set out separately, as Arbitration Format Theory, in a companion paper. It does not propose an alternative physical basis for consciousness, and its claims are stated over observables rather than over experience. Whether it is right is a separate question from whether the columns are single-occupant. They are so either way, and the experiment that tests them is worth running whoever turns out to have predicted the answer.

Supplementary Materials

Supplementary Materials: Tables S1–S3, the second-coder codebook and the inclusion criteria are deposited at [repository] under DOI [10.xxxx/PLACEHOLDER] and accompany the submission. Table S1: theory-by-quantity coding matrix (72 cells, with per-cell justification and source DOI); Table S2: content inventory (36 studies, with paradigm, manipulated content and theory attribution, and the inclusion rule for “theory-addressed”); Table S3: coding of the same theory set against the ConTraSt dimensions.

Data Availability Statement

Data Availability Statement: Tables S1–S3, the second-coder codebook and the inclusion criteria are deposited at [repository] under DOI [10.xxxx/PLACEHOLDER] and accompany the submission. No new empirical data were generated.

Declarations

Author Contributions: Conceptualisation, [Author 1]; methodology, [Author 1]; investigation (first coding of Tables S1–S3), [Author 1]; validation (blind second coding of Tables S1–S3 against the codebook), [Author 2]; writing – original draft, [Author 1]; writing – review and editing, [Author 1] and [Author 2]. Both authors have read and agreed to the published version of the manuscript.

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